

On the Decidability of $\mathcal{ALCI}_{\text{RCC5}}$ (and $\mathcal{ALCI}_{\text{RCC8}}$) Concept Satisfiability: Split-Forest Models, a Conditional Decidability Theorem, and a Machine-Checked Reduction to One Keystone

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Abstract

Whether concept satisfiability in $\mathcal{ALCI}_{\text{RCC5}}$ and $\mathcal{ALCI}_{\text{RCC8}}$ —the description logic $\mathcal{ALCI}$ with the RCC5/RCC8 base relations as roles under *abstract composition-table* semantics—is decidable has been open since Wessel’s 2002/2003 work. This paper reports a sustained, adversarially-reviewed, partly machine-checked attack on the problem, conducted by AI assistants under human direction. We do not settle the question. What we establish is: (i) a *conditional decidability theorem*—if the “live width” of models is bounded by a computable function of the concept (property F6), then satisfiability is decidable—whose soundness direction is *machine-checked in Lean 4*; (ii) that the entire remaining difficulty compresses into that single property F6, which we show is the *same fact* that blocks a proof of undecidability; (iii) a certified fragment of the local algebra (an RCC5 normal form) and an unconditionally decidable fragment (the $\forall\text{PO}$ -free concepts); and (iv) an honest map of the routes tried, the dead ends, and the missing hard pieces. We also report, candidly, on the methodology—adversarial “cold review,” machine verification, and a human-in-the-loop workflow—and on what frontier AI could and could not do.

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1 Introduction

1.1 AI disclaimer and authorship

The mathematical content of this paper was produced by AI assistants (Anthropic’s Claude models and OpenAI’s GPT-5.5), prompted, steered, and reviewed by the human author (Michael Wessel), who posed the problem, supplied the central intuitions and directions, orchestrated the review–repair cycles, and takes responsibility for the framing. **The results are unverified by human domain experts and should be read as AI-generated conjectures supported by machine checking and computational evidence—not as established theorems**, with the explicit exceptions of the parts stated to be kernel-checked in Lean 4 (the soundness pipeline, the abstraction’s faithfulness, and the RCC5 normal form), which are machine-verified but still human-unreviewed. We are deliberate about this labelling throughout: the standing one-line status is *strongly supported, not certified*. We regard the AI systems as having made contributions of a kind that, in a human collaborator, would warrant co-authorship, and we discuss this in §7.

1.2 Why the problem is hard, in one page

RCC5 relates two regions by exactly one of five relations: equal (EQ), proper part (PP), proper superpart (PPI), partial overlap (PO), or discreteness/disjointness (DR). Under *abstract composition-table* semantics a model is not a set of actual regions but a set with a total labelling of ordered pairs by these relations, required only to be converse-closed and *composition-closed*: if $x R y$ and $y S z$ then $x T z$ for some T the composition table allows.

The five relations split into two axes that behave completely differently.

- **The vertical axis** (PP, PPI: nesting) is *tame*. Composition can *force* vertical structure—the only four “determining” compositions (those returning a single relation) all run through PP or PPI—but an unbounded vertical chain folds into a finite loop, exactly as in ordinary modal logic. Infinite depth is not the problem.

- **The horizontal axis** (DR, PO: side-by-side) is *wild*. Composition can never *determine* a horizontal relation (PO is never a forced value), yet it can force two occurrences to be *distinct* (a shared horizontal anchor excludes EQ). So horizontal “crowds” of pairwise-distinct occurrences are the danger.

A finite decision procedure must record, at each interface of a model, the genuine branching it cannot reconstruct—the *live width*. Many forced edges are *shadows* (implied by composition from edges already present) and cost nothing; live width counts only the rest. The whole decidability question reduces to one property:

(F6) Is the live width of models bounded by a computable function of the input concept?

If yes, the certificate space is finite and satisfiability is decidable. And here is the knife’s edge: F6 is *the same fact* that blocks proving the problem *undecidable*. To force undecidability one would encode a rigid coordinate grid, which requires forcing a horizontal relation to a single value—exactly what the composition table cannot do (the “coincidence obstruction” Wessel identified in 2002/2003). Turn that looseness into a bound and one gets decidability; circumvent it and force a grid and one gets undecidability. The two open directions are one question seen from two sides, which is why neither has been settled in over twenty years.

1.3 Scope, contributions, and structure

This is both a technical report and a discussion piece. Its contributions are:

1. A *conditional decidability theorem* ($F6 \Rightarrow \text{decidable}$) whose soundness half is machine-checked in Lean 4 (§5.5), and the reduction of the whole problem to F6.
2. The identification of F6 as the shared keystone of the decidability and undecidability directions (§1.2, §8).
3. A certified RCC5 *normal form* and an unconditionally decidable $\forall\text{PO-free}$ fragment (§5.2, §5.6).
4. An honest, technique-organized account of every route tried, the counterexamples that killed the wrong ideas, and the lessons (§5), so that others need not repeat them.
5. A candid methodology report on adversarial AI-assisted mathematics (§7).

The paper is organized as follows. §2 states, in largely non-technical terms, where the problem stands after twenty-five years. §3 gives the history ($\mathcal{ALCC}\mathcal{I}_{\text{RCC}}$ from Cohn 1993 through Wessel 2002/2003 and Lutz–Wolter 2006) and carefully distinguishes our object from the neighbouring logics that *are* known undecidable. §4 explains why the standard undecidability reductions fail. §5—the core of the paper—surveys the approaches *by technique*: naive tableaux and blocking, finite-model and reduction attempts, the split-forest idea, the automata route, the Lean line culminating in F6, and the regular-cover pivot. §6 describes the working reasoners and a GIS example. §7 reports on the AI methodology. §8 concludes with what was learned, what would move the needle, an honest estimate of the chances of closing F6, and the project’s actual wins.

2 Current status: where we stand after twenty-five years

Decidability of $\mathcal{ALCT}_{\text{RCC5}}$ (and $\mathcal{ALCT}_{\text{RCC8}}$) concept satisfiability remains open. A current-literature check (July 2026) found no published resolution; the closest neighbours are decided in the other direction and are discussed in §3. What the present work adds is not an answer but a sharp reduction and a machine-checked surround.

The positive result—conditional decidability. *If the live width of models is bounded by a computable function of the concept (property F6), then concept satisfiability is decidable; and the soundness half of that reduction is machine-checked in Lean 4* (a valid finite certificate provably unfolds to a genuine RCC5 model of the concept, and the finite type abstraction is faithful). What is open is the antecedent F6 itself—equivalently the completeness direction, that every satisfiable concept has a *bounded* finite certificate.

What is certified (machine-checked, Lean 4, zero sorry):

- the *soundness* pipeline: a valid finite certificate unfolds to a genuine RCC5 model of the concept;
- the *faithfulness* of the finite (Hintikka) abstraction: satisfiability and abstraction-realizability coincide;
- the *RCC5 normal form* (forward): every strong-EQ RCC5 network is an ordered-disjoint structure—a strict partial order for PP with a downward-closed disjointness for DR, PO residual.

What is strongly supported but not certified: F6, the single open keystone—now sharpened to a precise combinatorial dichotomy (does some concept force unbounded “identity-selector minors”?), with the auxiliary uniformization property W2' shown to *fold into* it.

Unconditional by-products (independent of F6):

- the $\forall\text{PO}$ -free fragment is *decidable*—concepts with no $\forall\text{PO}$ subformula, still retaining $\exists\text{PO}$, $\forall\text{DR}$, $\exists\text{DR}$, and all vertical modalities, so a genuinely expressive spatial fragment;
- the certified RCC5 normal form above;
- working reasoners (a cover-tree tableau) that reproduce, 23 years later, the GIS taxonomy of Wessel’s original report;
- the standard undecidability routes shown blocked, with the coincidence obstruction identified as the shared fact.

The honest bottom line. After roughly thirty repair rounds and fifteen adversarial reviews the project has reached a genuine *local optimum*: the local algebra is exhausted (and in one case certified), the soundness side is machine-checked, and the whole remaining difficulty is one well-posed lemma. We offer the work not as a resolution but as a two-decade open problem *reduced, honestly and checkably, to its irreducible core*.

3 History of $\mathcal{ALCI}_{\text{RCC}}$ and why the known results do not settle it

3.1 Cohn’s 1993 proposal

The idea of combining description logics with qualitative spatial reasoning via modal accessibility relations was first proposed by Cohn [1], who considered treating the RCC8 relations (or subsets thereof) as modalities in a multi-modal logic. The framework allowed spatial constraints to be expressed as modal formulas—e.g., $\langle \text{PP} \rangle C$ for “some proper part satisfies C ”—but Cohn left all decidability and complexity questions open.

3.2 Wessel’s $\mathcal{ALCI}_{\text{RCC}}$ family (2002/2003)

Wessel [7, 8, 5] gave the first rigorous treatment of the combination, introducing the $\mathcal{ALCI}_{\text{RCC}}$ family: description logics $\mathcal{ALCI}_{\text{RCC}k}$ ($k = 1, 2, 3, 5, 8$) extending $\mathcal{ALCI}$ with role boxes derived from the RCC composition tables. In an $\mathcal{ALCI}_{\text{RCC}k}$ interpretation, every pair of domain elements is assigned exactly one RCC k base relation, and the composition table is enforced globally (*strong EQ semantics*: EQ is identity). Models are complete graphs with RCC k edge-colorings.

Wessel established the following:

- **Decidable cases:** $\mathcal{ALCI}_{\text{RCC}1}$, $\mathcal{ALCI}_{\text{RCC}2}$, and $\mathcal{ALCI}_{\text{RCC}3}$ are decidable, because their composition tables are deterministic or near-deterministic, admitting reduction to standard DL techniques.
- **Lower bounds:** $\mathcal{ALCI}_{\text{RCC}5}$ is PSPACE-hard; $\mathcal{ALCI}_{\text{RCC}8}$ is EXPTIME-hard.
- **Open:** The decidability of $\mathcal{ALCI}_{\text{RCC}5}$ and $\mathcal{ALCI}_{\text{RCC}8}$ was left as the central open problem.

In a series of companion reports, Wessel also mapped out the decidability landscape for description logics with general composition-based role inclusion axioms:

- $\mathcal{ALC}_{\mathcal{RA}}$ (arbitrary role axioms) is **undecidable**, via reduction from the Post Correspondence Problem [6].
- $\mathcal{ALC}_{\mathcal{RA}}^{\ominus}$ (non-disjoint-roles variant) is also **undecidable**, via reduction from non-empty CFG intersection [3].
- $\mathcal{ALC}_{\mathcal{RASG}}$ (admissible role boxes: deterministic, functional, complete, and associative) is **decidable** (EXPTIME-complete, via reduction to $\mathcal{ALC}$ with general TBoxes) [4]; adding unqualified number restrictions ($\mathcal{ALCN}_{\mathcal{RASG}}$) makes it undecidable again.

$\mathcal{ALCI}_{\text{RCC}5}$ sits precisely in the gap: its role box is non-deterministic (unlike $\mathcal{ALC}_{\mathcal{RASG}}$) but has the patchwork property (unlike $\mathcal{ALC}_{\mathcal{RA}}/\mathcal{ALC}_{\mathcal{RA}}^{\ominus}$).

Remark 1 (Key difficulties). $\mathcal{ALCI}_{\text{RCC}5}$ has none of the properties that standard DL decidability proofs rely on: (i) no tree model property (models are complete graphs), (ii) no finite model property (some satisfiable concepts require infinite models), (iii) a non-deterministic role box (multi-valued composition entries).

For reference, the full RCC5 composition table is given in Table 1. Reading: row = $S(b, c)$, column = $R(a, b)$, entry = possible relations for (a, c) .

Key composition facts exploited in the cover-tree theory (Sections ??–??): $\text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\}$ and $\text{comp}(\text{DR}, \text{PPI}) = \{\text{DR}\}$ (DR rigidity); $\text{comp}(\text{PP}, \text{PP}) = \{\text{PP}\}$ and $\text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$ (transitivity); and $\text{comp}(\text{DR}, \text{PO}) = \{\text{DR}, \text{PO}, \text{PP}\}$ (PP can be *forced* by composition and universal filtering; see Example ??).

Table 1: The RCC5 composition table (* = all five relations).

comp	DR(a, b)	PO(a, b)	EQ(a, b)	PPI(a, b)	PP(a, b)
DR(b, c)	*	DR, PO, PPI	DR	DR, PO, PPI	DR
PO(b, c)	DR, PO, PP	*	PO	PO, PPI	DR, PO, PP
EQ(b, c)	DR	PO	EQ	PPI	PP
PP(b, c)	DR, PO, PP	PO, PP	PP	PO, EQ, PP, PPI	PP
PPI(b, c)	DR	DR, PO, PPI	PPI	PPI	*

3.3 Wessel’s grid constructions and the coincidence obstruction

In his investigation of $\mathcal{ALCI}_{RCC8}$ [8], Wessel made two important discoveries:

1. Even-odd chains (Section 4.4.2). The TPP/NTPP distinction in RCC8 allows the construction of *functional successor chains*: each node alternates between *even* and *odd*, and a node can distinguish its direct TPPI-successor from all indirect ones by parity. Wessel used this to encode a binary n -bit counter, proving $\mathcal{ALCI}_{RCC8}$ is EXPTIME-hard.

2. The grid coincidence obstruction (Section 4.4.3, Figure 11). Wessel constructed explicit RCC8 networks isomorphic to $n \times n$ grids, using TPP for horizontal and vertical successors. However, he identified a fundamental obstacle to encoding domino tilings: the *coincidence condition* $R_X \circ R_Y = R_Y \circ R_X$ (east-of-north equals north-of-east) cannot be enforced by concept terms. The composition table allows $\{PO, EQ, TPP, NTPP, TPPI, NTPPI\}$ at the coincidence point—too many options. Wessel concluded:

“Even though these infinite models do exist, we have found no way to *enforce* the depicted model(s) by means of a concept term.”

Wessel further observed that adding a hybrid logic binding operator $\downarrow$ (nominals) would immediately yield undecidability, confirming that the coincidence condition is the *precise boundary*.

Wessel thus independently developed, three to four years before Lutz–Wolter, the RCC8 description-logic analogues of these ingredients—the even-odd chain, an explicit two-dimensional TPP/NTPP grid, and the first identification of the coincidence gap as the decisive obstacle. Lutz–Wolter later obtained undecidability by a *different* route—a linearized tiling enumeration (their §5, following Marx–Reynolds) that sidesteps the two-dimensional coincidence rather than enforcing it—so this is shared lineage and prior identification, not a dependence of their proof on Wessel’s constructions.

3.4 Lutz & Wolter (2006)

Lutz and Wolter [9] studied *modal logics of topological relations*, denoted L_{RCC8} , L_{RCC5} , etc. These have the same *syntax* as $\mathcal{ALCI}_{RCC}$ but are interpreted over *topological models*: the domain elements are regular closed subsets of a topological space (typically $\mathbb{R}^n$). They proved:

- L_{RCC8} is **undecidable** (their §5: a linearized NTPP-chain enumeration of tile positions, following Marx–Reynolds, whose faithfulness relies on the geometry of $\mathbb{R}^n$; no two-dimensional grid coincidence is enforced).
- $L_{RCC5}(RS^\exists)$ is **undecidable** (reduction from $S5^\exists$ via supremum regions).

Table 2: Why standard undecidability reductions fail for $\mathcal{ALCI}_{RCC5}$.

Source lem	Prob-	Technique	Missing $\mathcal{ALCI}_{RCC5}$	in
$\mathbb{N} \times \mathbb{N}$ domino		Graded modalities + transitivity	Number restrictions	
$\mathcal{ALC}_{\mathcal{RA}}$ [6]		Post Correspondence Problem	Arbitrary role axioms	
$\mathcal{ALC}_{\mathcal{RA}}^{\ominus}$ [3]		Non-empty CFG intersection	Non-disjoint roles	
$\mathcal{ALCI}_{RCC8}$ (Wessel)		Grid via TPP/NTPP	TPP/NTPP distinction	
L_{RCC8} (Lutz–Wolter)		Linearized topological enumeration	Topological rigidity	
$L_{RCC5}(RS^{\exists})$		$S5^3$ via supremum regions	Supremum closure	

- $L_{RCC5}(RS)$ is **open**. Lutz and Wolter explicitly wrote (p. 31): “Perhaps the most interesting candidate is $L_{RCC5}(RS)$ [...] to which the reduction exhibited in Section 8 does not apply.”

The logic $\mathcal{ALCI}_{RCC5}$ is exactly $L_{RCC5}(RS)$: arbitrary complete-graph models with composition-table semantics. The problem has been recognized as open by leading researchers since 2006.

4 Why standard undecidability reductions fail

Undecidability proofs for description logics use a variety of techniques: grid encoding ($\mathbb{N} \times \mathbb{N}$ domino tiling, requiring functional roles or number restrictions), reduction from the Post Correspondence Problem (PCP), reduction from context-free grammar (CFG) intersection, or simulation of $S5^3$ via geometric constructions. $\mathcal{ALCI}_{RCC5}$ lacks the features that each of these techniques exploits: it has no functional roles, no number restrictions, no arbitrary role box, and no geometric coincidence conditions.

4.1 Blocked reduction routes

Table 2 summarizes the blocked routes. The **patchwork property** of RCC5 (Renz & Nebel [2])—local (triple-wise) consistency implies global consistency for atomic networks—actively resists grid encoding, since tiling reductions require rigid global constraints that go beyond local consistency.

4.2 A concrete domino attempt in $\mathcal{ALCI}_{RCC5}$

Can we encode a domino-like grid using PP/PPI chains in RCC5? Consider the concept:

$$\begin{aligned}
X = & \exists PPI. \exists PPI. C \sqcap \exists PPI. \exists PPI. D \\
& \sqcap \forall PPI. \forall PPI. (\forall PO. (\neg C \sqcap \neg D) \sqcap \forall DR. (\neg C \sqcap \neg D)) \\
& \qquad \sqcap \forall PPI. (\neg C \sqcap \neg D) \sqcap \forall PP. (\neg C \sqcap \neg D)) \\
& \sqcap \forall PPI. \exists PPI. X
\end{aligned}$$

The intent: each X -node has two PPI-grandchildren colored C and D (the “square”), and the universals isolate the colored nodes from all non-EQ neighbors. The recursive $\forall\text{PPI}.\exists\text{PPI}.X$ forces infinite repetition.

One level is satisfiable. Without the recursive conjunct, X has a 3-element model: $z \text{ PP } y \text{ PP } x$, with $z \in C \sqcap D$. The two grandchildren are forced to be *the same element* (EQ collapse): if $z_1 \neq z_2$, they would be related by some $R \in \{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$, and the universals would require $z_2 \notin D$ (firing z_1 ’s $\forall R. \neg C \sqcap \neg D$). So $z_1 = z_2 \in C \sqcap D$, and the universals are vacuously satisfied (the only neighbors of z are PP-predecessors, not PO/DR/PPI-neighbors).

Two levels is unsatisfiable. Adding $\forall\text{PPI}.X$: the PPI-child y must itself satisfy X (without recursion), producing a new $C \sqcap D$ -labeled element w as a PPI-child of z . But z is a PPI-grandchild of x , and x ’s universals fire: z ’s PPI-neighbor w must satisfy $\neg C \sqcap \neg D$. Yet $w \in C \sqcap D$. **Contradiction.**

This was verified computationally: a simplified version $X_1 = \exists\text{PPI}.\exists\text{PPI}.C \sqcap \forall\text{PPI}^3.\neg C$ is SAT; $X_2 = X_1 \sqcap \forall\text{PPI}.X_1$ is UNSAT. The domino encoding collapses at depth 2 because the isolation constraints from the grandparent level kill the colored elements created by the recursive unfolding.

Remark 2. The failure is not accidental. In $\mathcal{ALCI}_{\text{RCC5}}$, the only way to create “functional” successors is via PP/PPI chains, but $\text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$ makes these chains *transitive*—grandparent universals automatically propagate to grandchildren, preventing the positional diversity needed for grid encoding. In $\mathcal{ALCI}_{\text{RCC8}}$, the TPP/NTPP distinction breaks this transitivity (a direct TPPI-child is distinguishable from indirect descendants), which is why Wessel’s even-odd chain works there but has no RCC5 analogue.

4.3 A concrete domino attempt in $\mathcal{ALCI}_{\text{RCC8}}$

The remark above suggests the TPPI/NTPPI distinction in $\mathcal{ALCI}_{\text{RCC8}}$ might rescue the construction: a direct successor (TPPI) and an indirect one (NTPPI) are genuinely distinguishable, so perhaps the level-2 collapse can be avoided. We attempt the analogue and show that while it escapes the RCC5 failure mode, it fails for a different reason—Wessel’s *coincidence obstruction* [8].

The schema. Let Corner be the concept that isolates a node from all non-EQ neighbors:

$$\text{Corner} = \bigwedge_{R \in \{\text{DC}, \text{EC}, \text{PO}, \text{TPP}, \text{NTPP}, \text{TPPI}, \text{NTPPI}\}} \forall R. (\neg C \sqcap \neg D).$$

The isolation is placed *locally on the colored grandchildren* rather than as an ancestral $\forall\text{TPPI}.\forall\text{TPPI}$ universal, so it cannot propagate transitively through the recursion:

$$\begin{aligned} X = & \exists\text{TPPI}.\exists\text{TPPI}.(C \sqcap \text{Corner}) \sqcap \exists\text{TPPI}.\exists\text{TPPI}.(D \sqcap \text{Corner}) \\ & \sqcap \forall\text{TPPI}.(\neg C \sqcap \neg D) \sqcap \forall\text{TPPI}.X. \end{aligned}$$

The third conjunct $\forall\text{TPPI}.(\neg C \sqcap \neg D)$ forces the corner relation $x \rightarrow z$ to be NTPPI (not TPPI), using $\text{comp}(\text{TPPI}, \text{TPPI}) = \{\text{TPPI}, \text{NTPPI}\}$.

Level 1 is satisfiable, with a real direct/indirect distinction. Without recursion, x has TPPI-children y_1, y_2 (the two “direct” positions) and a single corner z reached by TPPI-TPPI chains from both y_1 and y_2 . The EQ-collapse at level 2 works exactly as in RCC5: $z_1 \in C \sqcap \text{Corner}$ and $z_2 \in D \sqcap \text{Corner}$ are mutual non-EQ neighbors only if Corner is violated, so under strong-EQ semantics $z_1 = z_2 = z$. Crucially, $z \neq y_1, y_2$ in general: z is NTPPI of x while y_1, y_2 are TPPI—the direct/indirect distinction survives, which is a genuine RCC8 refinement over the RCC5 attempt of §4.2.

Level 2 collapses via the coincidence obstruction. Adding $\forall \text{TPPI}.X$: the TPPI-child y of x satisfies X , producing its own corner $w \in C \sqcap D \sqcap \text{Corner}$ at NTPPI-distance from y , and hence at NTPPI-distance from x via $\text{comp}(\text{TPPI}, \text{NTPPI}) = \{\text{NTPPI}\}$. Now x ’s own corner z and y ’s corner w are distinct grid positions in the intended reading—but as elements of the model they are related by some RCC8 base relation. Since z carries **Corner**, every non-EQ neighbor of z lies in $\neg C \sqcap \neg D$. But $w \in C \sqcap D$. Therefore $z = w$ under strong-EQ. **All corners across the grid coincide at a single point.**

Why the TPPI/NTPPI trick fails. In the RCC5 attempt the collapse is a *transitive-propagation* failure: the grandparent universal fires on great-grandchildren because $\text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$. Moving the isolation to a local **Corner** marker and using the non-transitive piece $\text{comp}(\text{TPPI}, \text{TPPI}) = \{\text{TPPI}, \text{NTPPI}\}$ eliminates that specific pathway—the universal at the grandparent no longer reaches deeper nodes. But a *different* obstruction takes over: the composition table permits any of $\{\text{PO}, \text{EQ}, \text{TPP}, \text{NTPP}, \text{TPPI}, \text{NTPPI}\}$ between two corners at different intended grid coordinates, and nothing in the concept language rules out EQ. The **Corner** markers themselves then force the EQ-merge. This is exactly the obstruction Wessel identified in report7.pdf, Figure 11 (cf. §3.3): the composition table does not carry the rigidity needed to keep distinct grid cells distinct.

Remark 3. The failure mode is structurally different from the RCC5 one but traces back to the same root cause. Lutz–Wolter’s L_{RCC8} reduction *circumvents* the coincidence obstruction rather than enforcing it: they linearize the plane into a single NTPP-chain (the Marx–Reynolds enumeration) and recover the vertical direction from the enumeration, so no “north-of-east = east-of-north” coincidence has to be forced; the geometry of $\mathbb{R}^n$ only has to keep that linear encoding faithful [9]. No such rigidity is available in the abstract RCC8 semantics of $\mathcal{ALCI}_{\text{RCC8}}$: a pure composition table allows corner points to merge freely, and the TPPI/NTPPI distinction—while useful locally for a direct/indirect separation at level 2—cannot be chained to enforce an unbounded $\mathbb{N} \times \mathbb{N}$ grid. This matches the open-problem status of $\mathcal{ALCI}_{\text{RCC8}}$ decidability noted in [9] and further discussed in [8].

5 Overview of approaches, by technique

We organize the attack by *technique*, in roughly the order tried, not by round or date. Each subsection follows the same shape: the high-level idea; the concrete techniques investigated; where and why they fail (with the examples that made the failure clear); pointers to the papers, probes, and reasoners; and the lessons.

5.1 Naive tableaux and blocking

Idea. The first and most natural attack is a description-logic tableau: build a tree-shaped model top-down, expand each $\exists R.C$ by a fresh R -successor, propagate universals, and *block* a node when its type repeats an ancestor’s so the procedure terminates. The refined version we implemented is the *cover-tree tableau* (§6), which blocks on *exact occurrences* rather than mere types.

Techniques investigated. Because RCC5 assigns a relation to *every* pair, the model is a complete edge-coloured graph, and ordinary subset blocking is unsound. We investigated a ladder of stronger blocking conditions, each tracking more relational context: *triangle types* and triangle blocking (track composition triangles (x, y, z) ; `triangle_blocking_ALCIRCC5`, `src/triangle_*.py`), *quadruple types* (decidability via quadruples `ALCIRCC5`, `src/quadruple_type_check.py`), *profile blocking* (`src/profile_blocking_*.py`), *double blocking* (the *SHIQ* device for inverse roles;

`src/double_blocking_test.py`), and pairwise/neighbourhood blocking (`src/tri_neighborhood_check.py`, `src/three_arm_star_check.py`).

Why they fail. All of them founder on the same three phenomena, which are the reason the total-relation setting is genuinely harder than ordinary DL:

- **Composition forces edges *into* old nodes.** From $x \text{ PP } y$ and $y \text{ DR } z$, the table forces $x \text{ DR } z$: a new edge into the already-processed x . Such forced back- and cross-edges destroy the tree structure that blocking relies on and can add new obligations to a blocked node.
- **Dormant universals.** A $\forall R.C$ sits inert until an R -edge appears; a forced composition edge (above) can *later* fire a universal at a node already blocked, so the block was unsound. Re-qualifying an old node can *unblock* it, and the cascade need not terminate.
- **The one-point extension problem.** A configuration can be locally consistent (every triangle composition-closed) yet admit *no* extension by a required fresh witness—path-consistency is not extensibility.

The examples that made this clear. Two families were decisive. The forced-composition UNSAT concepts such as $C_{\text{force}} = \exists \text{PP}.(\exists \text{DR}.A) \sqcap \forall \text{DR}.\neg A$ (here $x \text{ PP } y, y \text{ DR } z \Rightarrow x \text{ DR } z$, so $\forall \text{DR}.\neg A$ kills the required $z:A$) show a naive procedure *accepting* unsatisfiable input if it ignores forced edges. And the strong-EQ PO-loop, $C \sqcap \exists \text{PO}.\exists \text{PO}.C \sqcap \forall \{\text{PO}, \text{DR}, \text{PP}, \text{PPI}\}.\neg C$, is satisfiable only via a two-element symmetric loop under EQ-identity — a model a type-based procedure *rejects*. These and further diagnostics (`inter_witness_counterexample_ALCIRCC5`, `typed_patchwork_counterexample_ALCIRCC5`) are exercised against the cover-tree tableau as the oracle.

Lesson. Sound termination in the total-relation setting cannot be obtained by local blocking, because the relational completion is global. One must *separate* the tree skeleton from the relational completion and control the latter by the patchwork property — which is exactly the split-forest idea (§5.3).

5.2 Finite-model and reduction attempts

Idea. If models can be forced finite (or finitely presented), one can enumerate and check. The concrete techniques: collapse infinite PP/PPI chains into finite *clusters/loops*; build a *quasimodel* and eliminate inconsistent types; take a *two-tier quotient* (one kernel per recurrent chain descriptor, plus finitely many designated witnesses); or *encode into MSO* over trees and invoke Rabin’s theorem.

Where they fail. The quasimodel / type-elimination route (`decidability_ALCIRCC5`, `src/alcircc5_reasoner`) is *retracted*: its Q3 anti-monotonicity causes cascade elimination and hence incompleteness (not unsoundness). The reasoner remains SAT-trustworthy but is provably incomplete on the PO-loop pattern above. The MSO route (`MSO_encoding_ALCIRCC5`) needs the horizontal crowd width to be MSO-definable/bounded — which is F6 again. Every “bound the model” variant reduces to producing a computable width bound, i.e. to F6.

The win. One fragment falls out unconditionally. In the two-tier quotient, PO is the only relation whose recurrence needs an all-phase safety condition; a concept with *no* $\forall \text{PO}$ *subformula* makes that condition vacuous, so it has a finite quotient and a computable live-width bound $K(C) =$

$(1+|cl(C)|)^2 2^{2^{cl(C)}}$. Hence the $\forall$ PO-free fragment is decidable (`two_tier_quotient_ALCIRCC5`, `po_free_fragment_ALCIRCC5`). It retains $\exists$ PO, $\forall$ DR, $\exists$ DR, and all vertical modalities, so it is a genuinely expressive spatial fragment — the strongest unconditional $\mathcal{ALCI}_{RCC5}$ -level result of the project.

Lesson. Finiteness is the wrong target: models are legitimately infinite (unbounded PP-towers). The right target is a finite *presentation* of a possibly-infinite model — which is the split-forest.

5.3 The split-forest idea

Idea (the human’s whiteboard). Decompose a model into a *forest* of trees — one per connected component — by splitting each join node into EQ-mates, so the tree skeleton is explicit and the (global) relational completion is handled separately by the patchwork property. Tree-like presentations of infinite models are the classical route to decidability (tree-model property, then automata or type-elimination).

The normal form (Theorem A). Every satisfiable concept admits a witness-generated, tree-like split-forest model, obtained by *witness thinning* $\rightarrow$ *generated containment covers* $\rightarrow$ *split copies* for incomparable superparts $\rightarrow$ *side-interface mosaics* $\rightarrow$ a *typed-EQ quotient* (§??; `trees/cover_tree_tableau_ALCIRCC5`). This is the one component found sound by *every* cold review — the semantic foundation everything downstream *consumes*.

Transition. The split forest gives the model structure; it does not by itself give a *decision procedure*. Turning “a tree-like model of this shape exists” into “an algorithm can decide whether one exists” is the finite-abstraction problem — and that is where the next two families (the certificate and the automaton) live.

5.4 The no-automata certificate, and the automata route

The hand certificate (rounds 2–10). A finite *certificate* for satisfiability over the split forest, validated by local checks (converse, composition, universal safety, request discharge, residual-frontier constraints; `gpt5.5_round2/`, `gpt5.5_round4/`). Across ten rounds it repeatedly reduced to automaton-shaped machinery it could not discharge by hand: request-closed cycles behave like Büchi/parity acceptance, and “finite product-state exhaustion” like product-automaton reachability. Cold reviews found successive defects — an ancestor-tower gap (D-1), its dual descendant tower, boundary non-determinacy, and (twice) that finite-checkability was asserted rather than proved. The thread converged, honestly, onto an automaton.

The parity-automaton route (rounds 11–12). This made the convergence explicit: Theorem A $\rightarrow$ Theorem B (a *finite abstraction* captures saturated split forests up to satisfiability) $\rightarrow$ Theorem C (a two-way alternating parity *tree* automaton accepts exactly the valid abstractions; emptiness is decidable). The automaton *consumes* the split forest rather than replacing it. Theorem B is the single load-bearing keystone. Cold reviews (5th, 6th) found its global-coherence step assumed, not proved; round 12 rebuilt the abstraction on *patchwork bags* so global consistency follows from the citable RCC5 patchwork property (Renz–Nebel), with eventualities decided by two-way emptiness (Vardi). A 7th (hot) review then found a residual *Shadow-Amalgamation* gap in the truth lemma (`automata_route_repairs/`, `cold_review_theoremB/`, probes `wp13-wp17`).

Why we abandoned the tree-automaton route. A cold review of the decision layer showed the tree automaton must decide an *existential-projection* language (the valid abstraction is a pair, only part of which is on the tape), and independent alternating run copies cannot be forced to guess the hidden part consistently — machine-exhibited as acceptance-without-validity. The three natural repairs (tape annotation, profile quotient, canonical guessing) are each blocked by facts the construction itself establishes. The decision layer was therefore replaced by a mosaic / type-elimination procedure directly on the infinitary abstraction — the cluster-quasimodel line (§5.5) — and the keystone (Theorem B $\approx$ patchwork-bag adequacy) survived the change, unmoved: it is F6 in automaton clothing.

5.5 The Lean line and F6

Lean, and how we use it. Lean 4 is a proof assistant whose kernel accepts a proof only if every step type-checks. In an adversarial project where prose proofs repeatedly *assumed* the hard step, the kernel is a ground-truth oracle: it cannot be talked past. We moved the load-bearing soundness argument into Lean 4 core (no `mathlib`); the development is `formal/Round19Transport.lean` ($\approx$ 5,300 lines, zero `sorry`), described in full in `LEAN.md`.

The cluster-quasimodel decision layer (rounds 15–18). A finite *catalogue* of interface patterns carrying *steering functions*, with one-step demand fulfilment (no promissory witnesses, hence no parity/eventuality condition). This is the construction on which the whole difficulty finally localized to two named lemmas: **F6** (a computable bound on *live width*) and **W2'** (uniformization of the extraction).

What is formalized (rounds 19–30). The pipeline from certificate syntax to a genuine RCC5 model of the logic is kernel-checked (transport $\rightarrow$ S-layer closure $\rightarrow$ catalogue soundness $\rightarrow$ logic layer / truth lemma $\rightarrow$ model), as is the *faithfulness* of the Hintikka abstraction (satisfiability and Hintikka-realizability coincide). On top sits a *non-oracular decision-grade reduction*: a fixed Boolean checker of first-order finite certificate data, proved sound, from which **Decidable** satisfiability follows by finite search modulo a single premise — exactly $F6 \wedge W2'$. This is the machine-checked *conditional decidability theorem*. Two cold reviews of the Lean confirmed no soundness defect and drove the closure of the completeness-interface gaps.

The keystone, sharpened. F6 is now a precise combinatorial dichotomy: does some concept force unbounded *identity-selector minors* (private matching of unbounded hitting number)? $W2'$ was shown false in its coarse form but repairable by a state refinement bounded *by F6* — so it *folds into* F6 (`fable5_width_barrier/`, `cold_review_f6_w2prime/`, probes `wp26-wp44`).

5.6 The regular-cover pivot

Idea. A reframing suggested by the human: replace finite *quotients* (invalid for infinite PP-chains, since a PP-cycle contradicts EQ-identity) by *regular covers* — finite generators whose infinite *unfoldings* are the models, in the spirit of the two-tier quotient but for the full logic. This finer invariant can finitely represent exactly the patterns that fool the width view (identity selectors $i = j$, half-graphs $i < j$).

What it produced. A complete *local* algebra of strong-EQ RCC5 networks, of which one piece is machine-certified (`formal/RCC5NormalForm.lean`): the RCC5 *normal form* — every strong-EQ RCC5 network is an *ordered-disjoint* structure (PP a strict partial order, DR a downward-closed disjointness, PO residual) — together with free amalgamation, an exact one-point extension criterion, and exact uniform/non-uniform cross-policy characterizations (`rcc5_local_amalgamation_theory`, `regular_cover_route_pivot/`, probes wp45–wp83). It independently re-derives the $\forall$ PO-free decidability result.

Why it is a better map, not a shorter path. Its own keystone (a bounded coherent regular-cover property) is F6 relabelled, and the route itself concedes, after exhausting the local algebra, that the remainder “cannot be a purely local RCC5 composition problem.” Because RCC5 relations are total, every model’s Gaifman graph is complete, so no off-the-shelf guarded/bounded-treewidth cover theorem applies; the keystone would need a bespoke cover theorem for complete edge-coloured regular structures. The pivot is the right vocabulary for a future attack, but it does not move the summit.

6 Current implementation: reasoners

7 On agentic AI usage

8 Summary and conclusion

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